

Progress Report 2

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Task 1

Task 1 – Static and Dynamic Loading Calculations for Critical Shaft

1.1 Critical Shaft Selection

The output shaft, S4, is selected as the critical shaft because it carries the winch drum load and has a 100 mm protruding section outside the gearbox. This creates the largest bending moment compared with the other shafts.

Material selected:

Aluminum 6063-O

Material properties:

$$S_y = 49MPa$$

$$S_{ut} = 110MPa$$

Shaft diameter:

$$d = 8mm = 0.008m$$

Output shaft speed:

$$N_{out} = 88rpm$$

1.2 Output Shaft Geometry

The internal bearing-to-bearing shaft length is:

$$L = 117mm = 0.117m$$

The 100T output gear is positioned at:

$$x_g = 37.5mm = 0.0375m$$

Bearing B is at:

$$x_B = 117mm = 0.117m$$

The output shaft protrudes 100 mm after Bearing B, so the drum load acts at:

$$\begin{aligned}x_{load} &= 117 + 100 = 217mm \\x_{load} &= 0.217m\end{aligned}$$

1.3 Nominal Output Torque

Load force:

$$\begin{aligned}F &= mg \\F &= (1)(9.81) = 9.81N\end{aligned}$$

Drum radius:

$$r = 12mm = 0.012m$$

Nominal output torque:

$$\begin{aligned}T &= Fr \\T &= (9.81)(0.012) = 0.118Nm\end{aligned}$$

So, the output shaft is analyzed at:

$$T = 0.118Nm$$

1.4 Gear Forces on Output Shaft

The output shaft carries the final 100T gear.

Pitch diameter of 100T gear:

$$d_g = 100mm = 0.100m$$

Tangential gear force:

$$F_t = \frac{2T}{d_g}$$

$$F_t = \frac{2(0.118)}{0.100} = 2.36N$$

For a 20° pressure angle:

$$F_r = F_t \tan (20^\circ)$$

$$F_r = 2.36 \tan (20^\circ) = 0.86N$$

Therefore:

$$F_t = 2.36N$$

$$F_r = 0.86N$$

The force diagram for the output shaft is shown below. The shaft is supported by Bearing A and Bearing B, with the 100T gear located at $x = 37.5mm$, Bearing B at $x = 117mm$, and the drum load acting at $x = 217mm$.

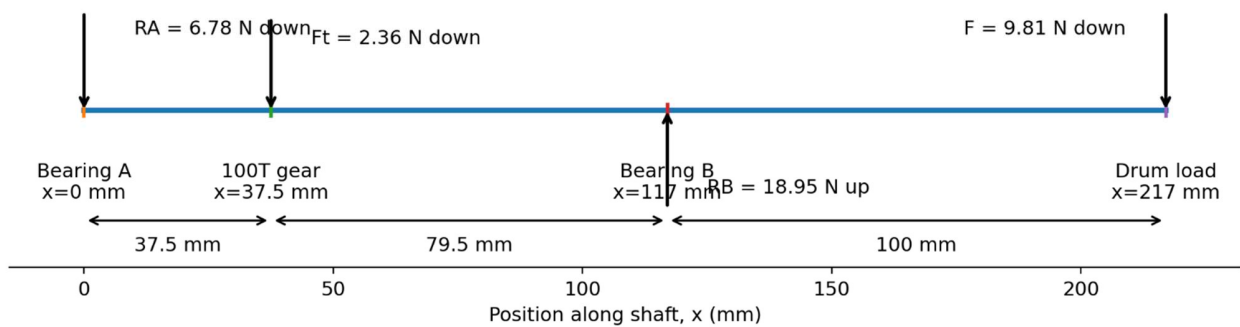


Figure 1. Force diagram of the critical output shaft showing the gear force, bearing reactions, and drum load.

1.5 Vertical Bearing Reactions

Vertical loads are the gear tangential force and the drum load.

$$R_A + R_B = 2.36 + 9.81$$

$$R_A + R_B = 12.17N$$

Taking moments about Bearing A:

$$R_B(0.117) = 2.36(0.0375) + 9.81(0.217)$$

$$R_B(0.117) = 0.0885 + 2.1288$$

$$R_B = \frac{2.2173}{0.117}$$

$$R_B = 18.95N$$

Then:

$$R_A = 12.17 - 18.95$$

$$R_A = -6.78N$$

The negative sign means Bearing A reacts downward.

1.6 Shear Force and Bending Moment

Position	Shear Force N	Bending Moment Nm
Bearing A, $x = 0$	-6.78	0
Gear, $x = 37.5mm$	-9.14after gear	-0.254
Bearing B, $x = 117mm$	+9.81after bearing	-0.981
Drum load, $x = 217mm$	0	0

Maximum bending moment occurs at Bearing B:

$$M_{max} = 9.81(0.100)$$

$$M_{max} = 0.981Nm$$

So the critical section is at Bearing B, where the overhanging output shaft begins.

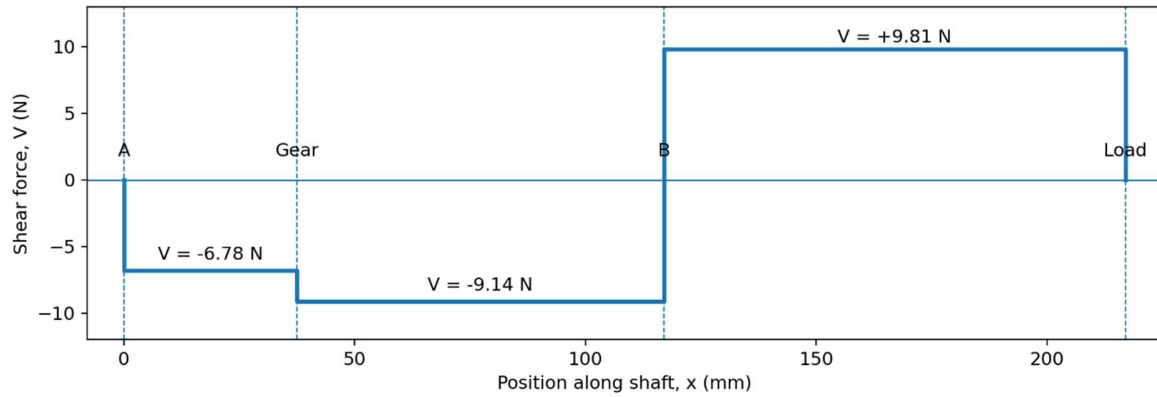


Figure 2. Shear force diagram for the critical output shaft.

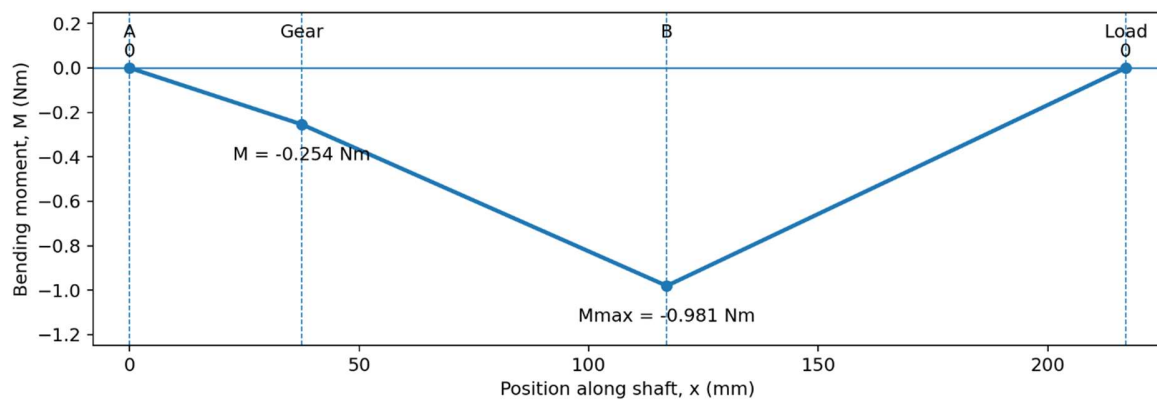


Figure 3. Bending moment diagram for the critical output shaft.

1.7 Horizontal Plane Check

The radial gear force also acts on the shaft:

$$F_r = 0.86N$$

Horizontal reaction at Bearing B:

$$R_{Bh} = \frac{0.86(0.0375)}{0.117}$$

$$R_{Bh} = 0.276N$$

Horizontal reaction at Bearing A:

$$R_{Ah} = 0.86 - 0.276$$

$$R_{Ah} = 0.584N$$

The horizontal bending moment is small compared with the vertical bending moment, so the critical bending moment remains:

$$M_{critical} = 0.981Nm$$

1.8 Static Stress Analysis

Bending Stress

$$\sigma_b = \frac{32M}{\pi d^3}$$

$$\sigma_b = \frac{32(0.981)}{\pi(0.008)^3}$$

$$\sigma_b = 19.52MPa$$

Torsional Shear Stress

$$\tau = \frac{16T}{\pi d^3}$$

$$\tau = \frac{16(0.118)}{\pi(0.008)^3}$$

$$\tau = 1.17MPa$$

Von-Mises Stress

$$\sigma_{vm} = \sqrt{\sigma_b^2 + 3\tau^2}$$

$$\sigma_{vm} = \sqrt{(19.52)^2 + 3(1.17)^2}$$

$$\sigma_{vm} = 19.62MPa$$

Yield Factor of Safety

$$n_y = \frac{S_y}{\sigma_{vm}}$$
$$n_y = \frac{49}{19.62}$$
$$n_y = 2.50$$

Since:

$$n_y > 1$$

the output shaft is safe against yielding.

1.9 Dynamic Loading Analysis

For a rotating shaft, the bending stress is treated as alternating. The torque is treated as steady mean torsion.

Alternating bending stress:

$$\sigma_a = 19.52MPa$$

Mean bending stress:

$$\sigma_m = 0$$

Alternating torsional stress:

$$\tau_a = 0$$

Mean torsional stress:

$$\tau_m = 1.17MPa$$

Equivalent alternating stress:

$$\begin{aligned}\sigma'_a &= \sqrt{\sigma_a^2 + 3\tau_a^2} \\ \sigma'_a &= 19.52 \text{ MPa}\end{aligned}$$

Equivalent mean stress:

$$\begin{aligned}\sigma'_m &= \sqrt{\sigma_m^2 + 3\tau_m^2} \\ \sigma'_m &= \sqrt{3(1.17)^2} \\ \sigma'_m &= 2.03 \text{ MPa}\end{aligned}$$

For aluminum:

$$\begin{aligned}S'_e &= 0.4S_{ut} \\ S'_e &= 0.4(110) = 44 \text{ MPa}\end{aligned}$$

Assuming machined/cold-drawn surface finish:

$$k_a = 1.0$$

Size factor for 8 mm shaft:

$$k_b = 0.995$$

Modified endurance strength:

$$\begin{aligned}S_e &= k_a k_b S'_e \\ S_e &= (1.0)(0.995)(44) \\ S_e &= 43.8 \text{ MPa}\end{aligned}$$

Using the Modified Goodman equation:

$$\begin{aligned}\frac{1}{n_f} &= \frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}} \\ \frac{1}{n_f} &= \frac{19.52}{43.8} + \frac{2.03}{110} \\ \frac{1}{n_f} &= 0.464 \\ n_f &= 2.15\end{aligned}$$

Since:

$$n_f > 1$$

The output shaft is safe against fatigue failure.

1.10 Critical Bearing Life Analysis

The critical bearing is Bearing B because it has the highest reaction force.

Vertical reaction at Bearing B:

$$R_B = 18.95N$$

Horizontal reaction at Bearing B:

$$R_{Bh} = 0.276N$$

Resultant bearing load:

$$\begin{aligned}P &= \sqrt{R_B^2 + R_{Bh}^2} \\ P &= \sqrt{(18.95)^2 + (0.276)^2} \\ P &= 18.95N\end{aligned}$$

A standard SKF 608 bearing has dimensions of 8 mm bore, 22 mm outside diameter, and 7 mm width, with basic dynamic load rating:

$$C = 3.45kN = 3450N$$

and static load rating:

$$C_0 = 1.37kN = 1370N$$

For ball bearings:

$$p = 3$$

Bearing life in million revolutions:

$$\begin{aligned} L_{10} &= \left(\frac{C}{P}\right)^p \\ L_{10} &= \left(\frac{3450}{18.95}\right)^3 \\ L_{10} &= 6.03 \times 10^6 \text{ million revolutions} \end{aligned}$$

Bearing life in hours:

$$\begin{aligned} L_{10h} &= \frac{L_{10} \times 10^6}{60N} \\ L_{10h} &= \frac{(6.03 \times 10^6)(10^6)}{60(88)} \\ L_{10h} &= 1.14 \times 10^9 h \end{aligned}$$

The bearing life equation uses $p = 3$ for ball bearings and converts million revolutions to operating hours using shaft speed.

For 98% reliability:

$$a_1 = 0.33$$

Adjusted bearing life:

$$\begin{aligned}L_{2h} &= a_1 L_{10h} \\L_{2h} &= 0.33(1.14 \times 10^9) \\L_{2h} &= 3.77 \times 10^8 h\end{aligned}$$

Required life:

$$15000h$$

Since:

$$3.77 \times 10^8 h > 15000h$$

The selected 608 bearing satisfies the bearing life requirement.

Static bearing check:

$$\begin{aligned}\frac{C_0}{P} &= \frac{1370}{18.95} \\ \frac{C_0}{P} &= 72.3\end{aligned}$$

Therefore, the bearing is also safe under static loading.

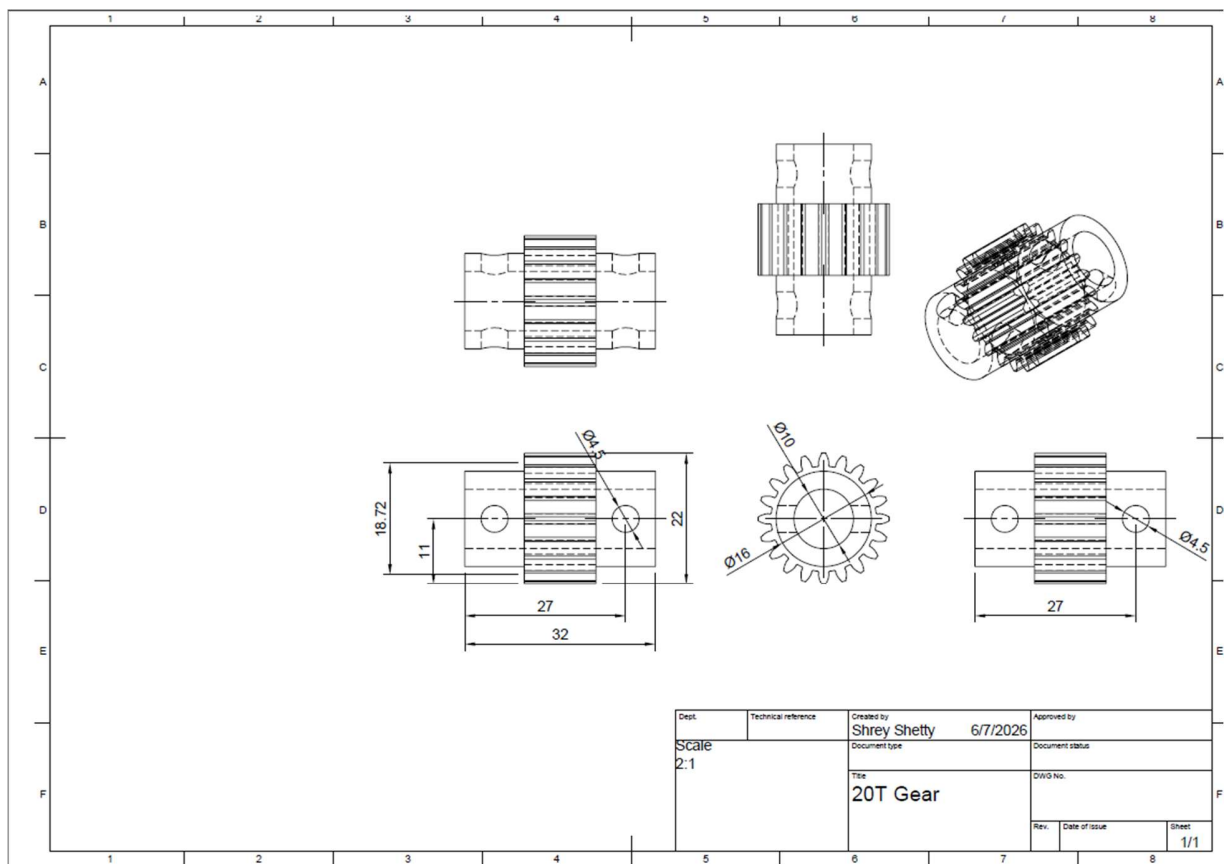
FINAL STATEMENT

The output shaft was identified as the critical shaft because it carries the winch drum load on a 100 mm protruding section outside the gearbox. Based on the static loading analysis, the maximum bending moment occurs at Bearing B and is 0.981 Nm. The calculated von-Mises stress is 19.62 MPa, resulting in a yield factor of safety of 2.50. Dynamic loading analysis using the Modified Goodman criterion produced a fatigue factor of safety of 2.15, indicating that the shaft is safe against fatigue failure. The critical bearing is Bearing B, with an adjusted life of (3.77×10^8) hours at 98% reliability, which is significantly greater than the required 15,000 hours. Therefore, both the output shaft and critical bearing satisfy the design requirements. The shaft analysis was performed using the output shaft geometry with key positions located at $(x = 0)$, (37.5) , (117) , and (217) mm.

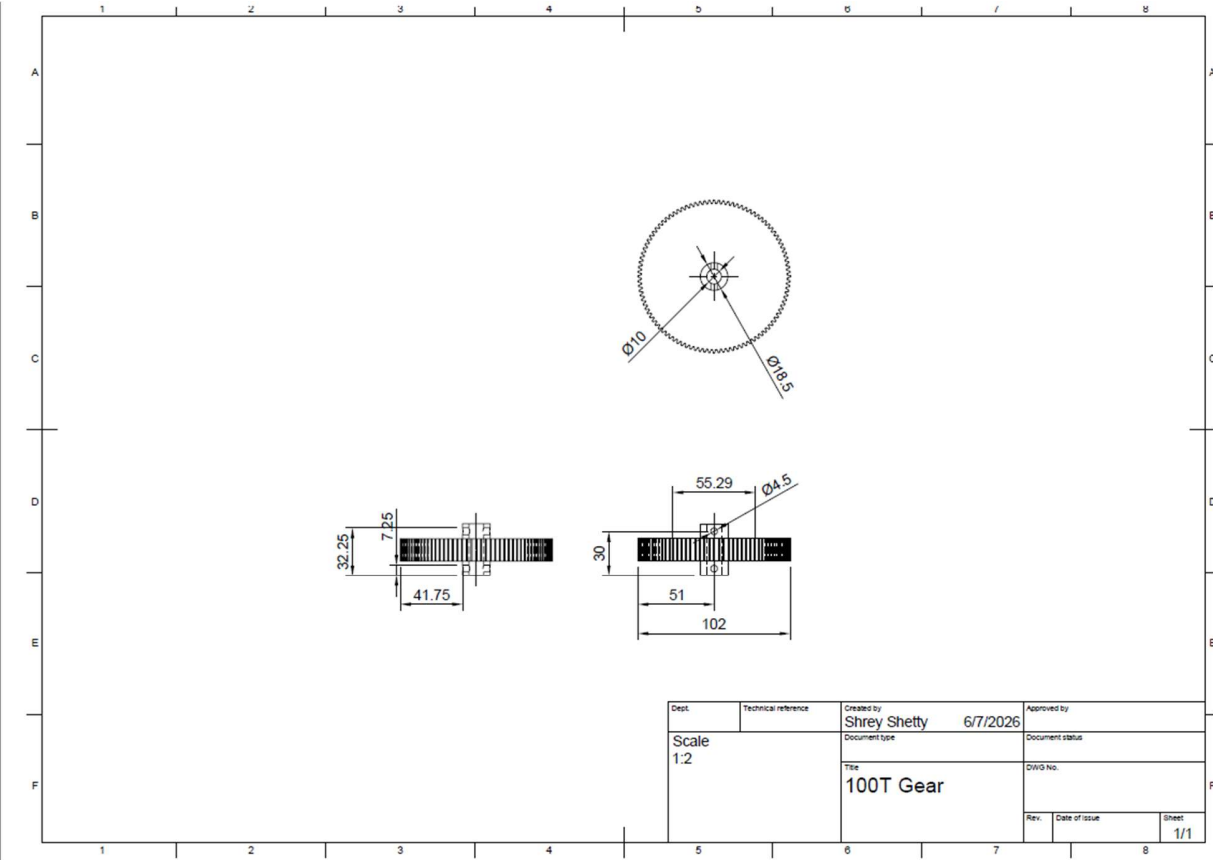
Task 2

Engineering / Assembly drawings

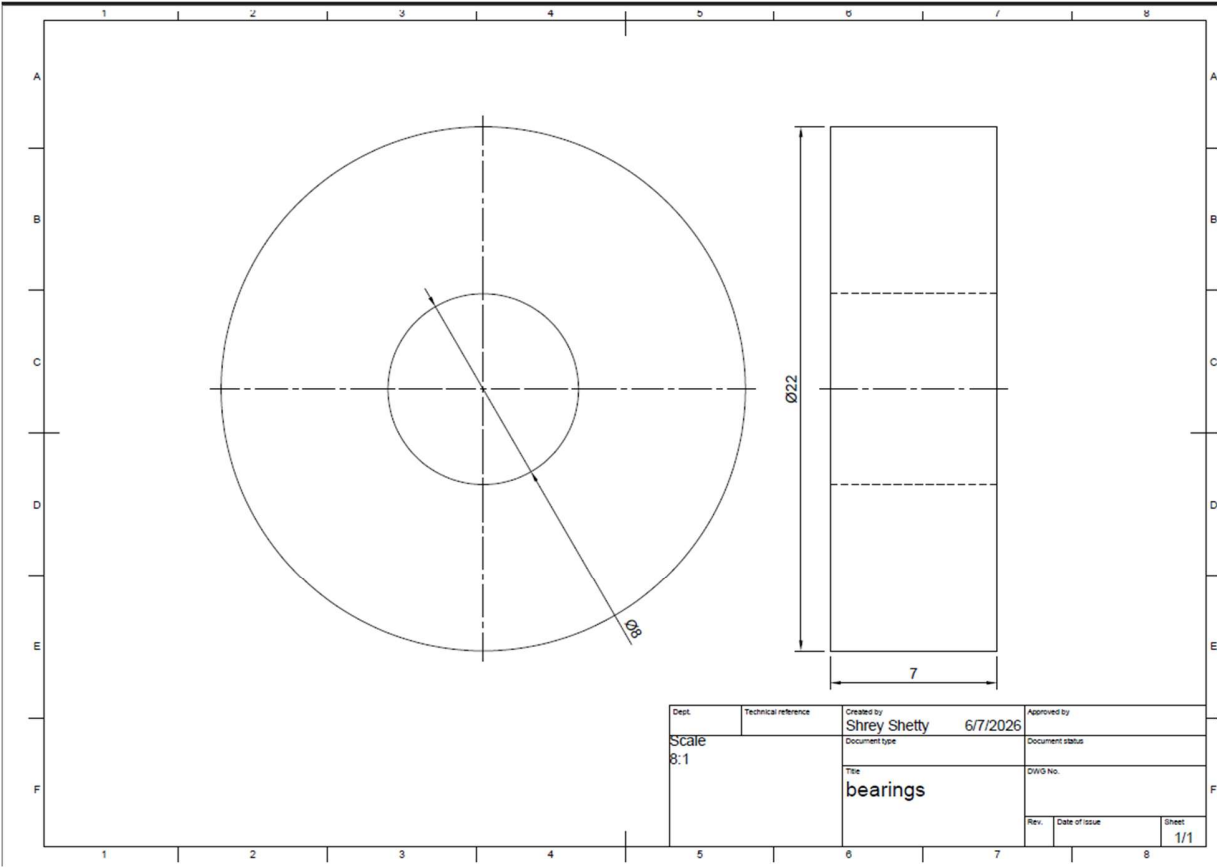
20 T Gear – PLA Plastic – 3D Printed



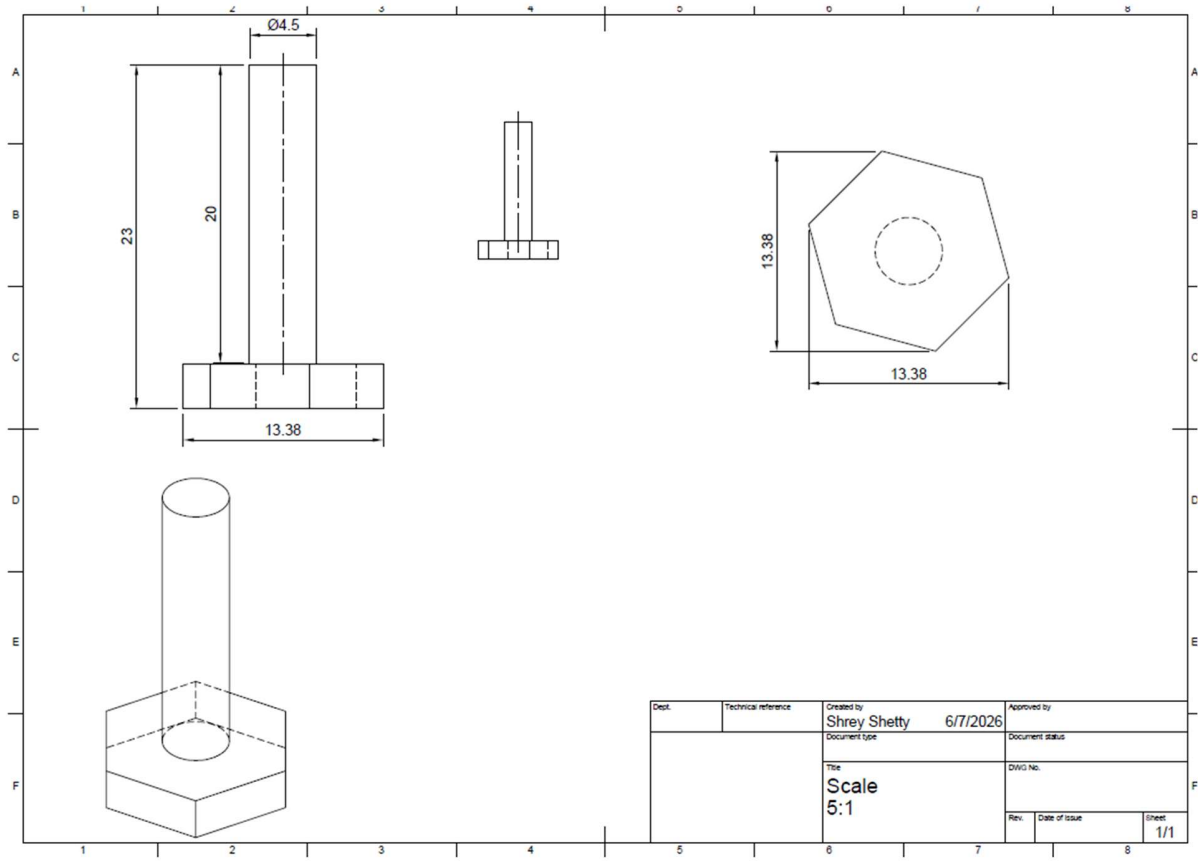
100 T Gear – PLA Plastic – 3D Printed



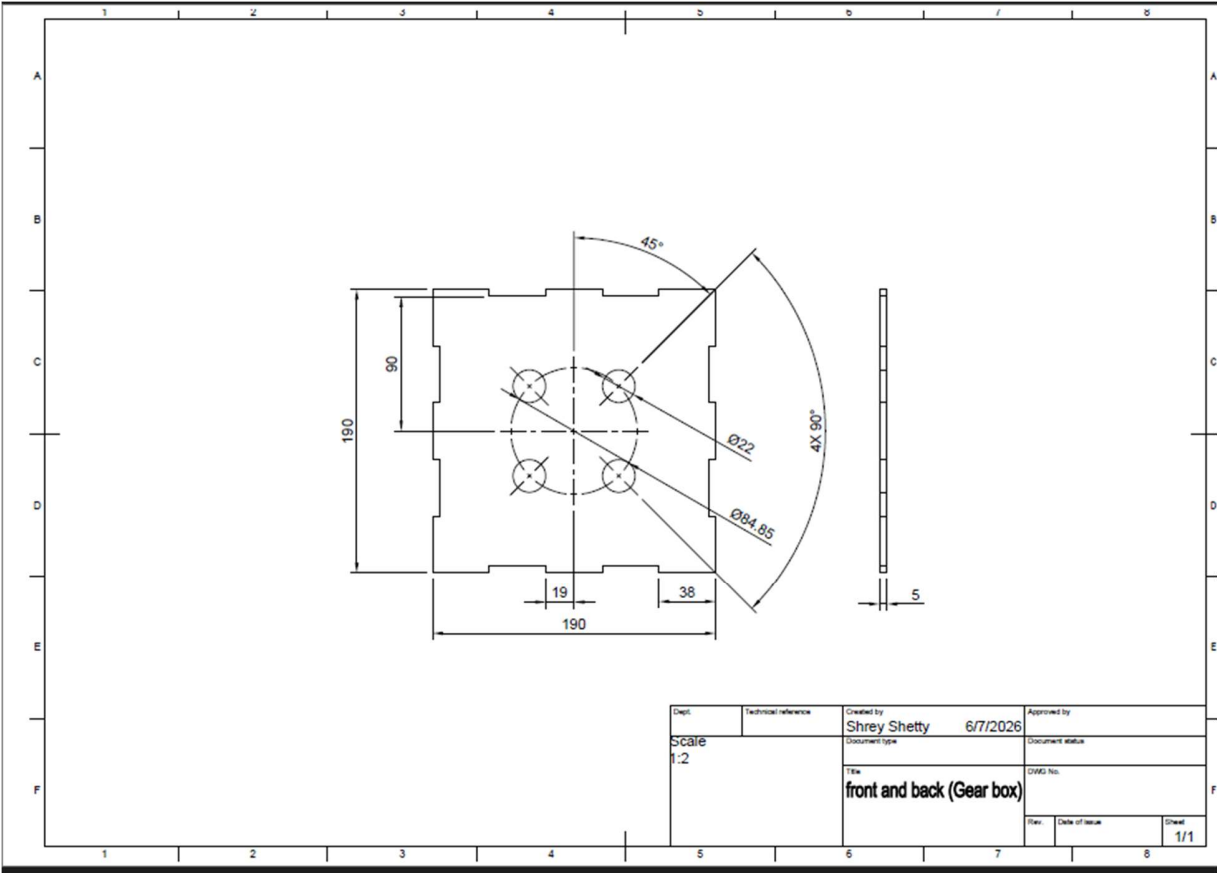
Bearing – Aluminum



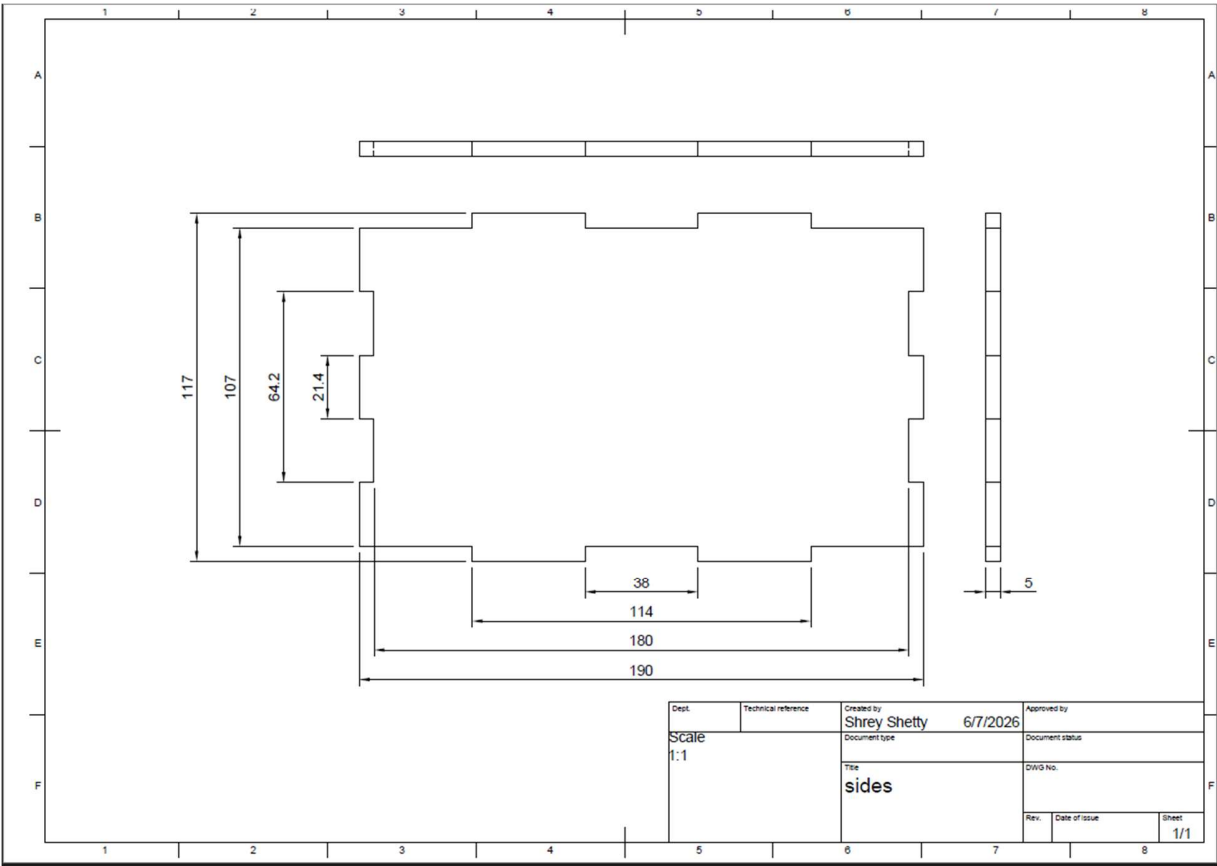
Bolt - Metal



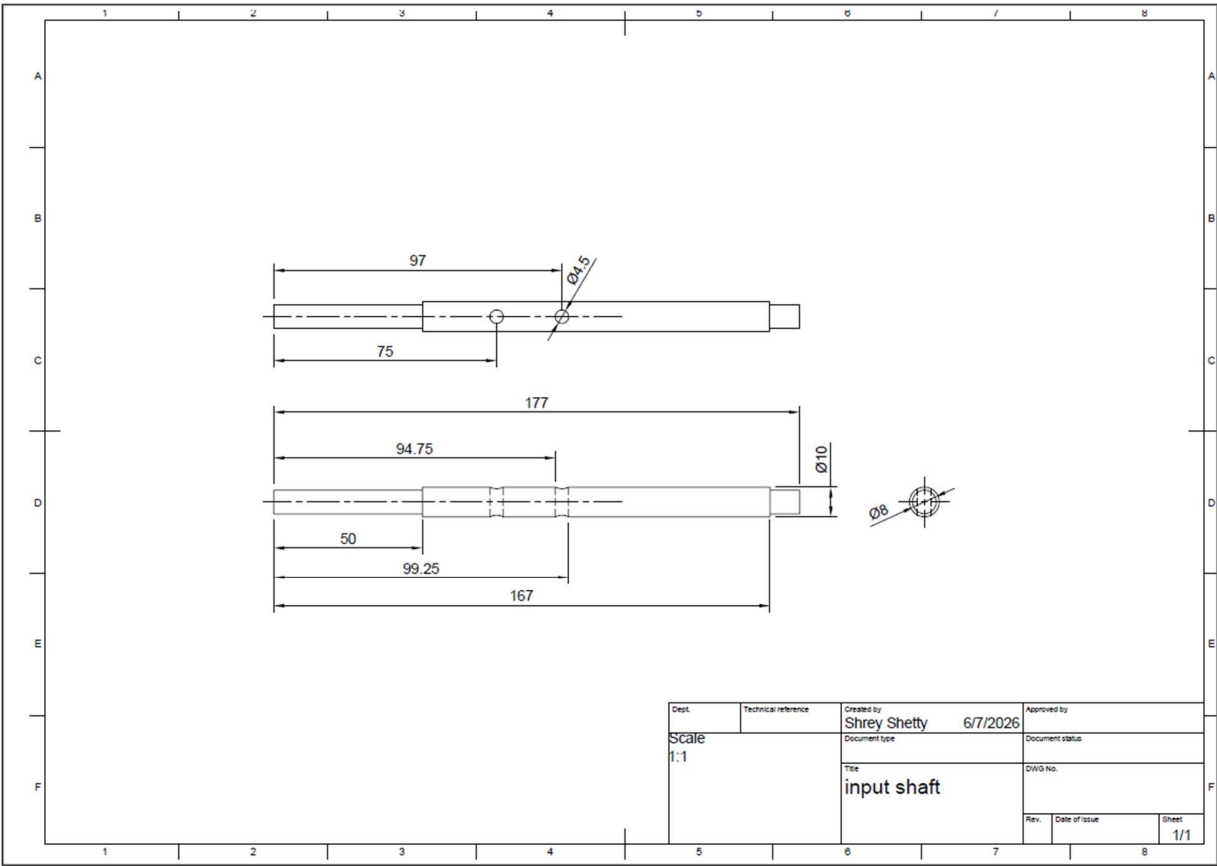
Front and Rear panel of Gear box housing – Acrylic – Laser cut



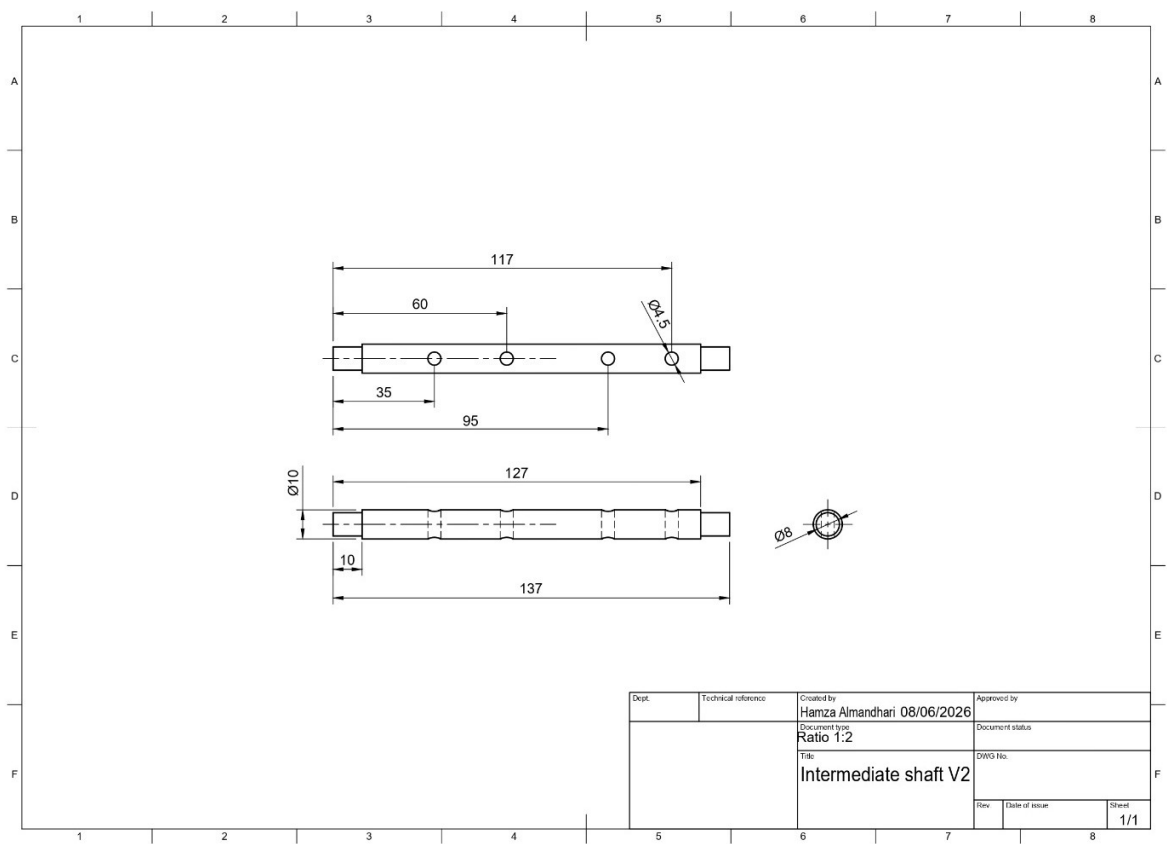
Side Panel of Gear Box housing – Acrylic - Laser cut



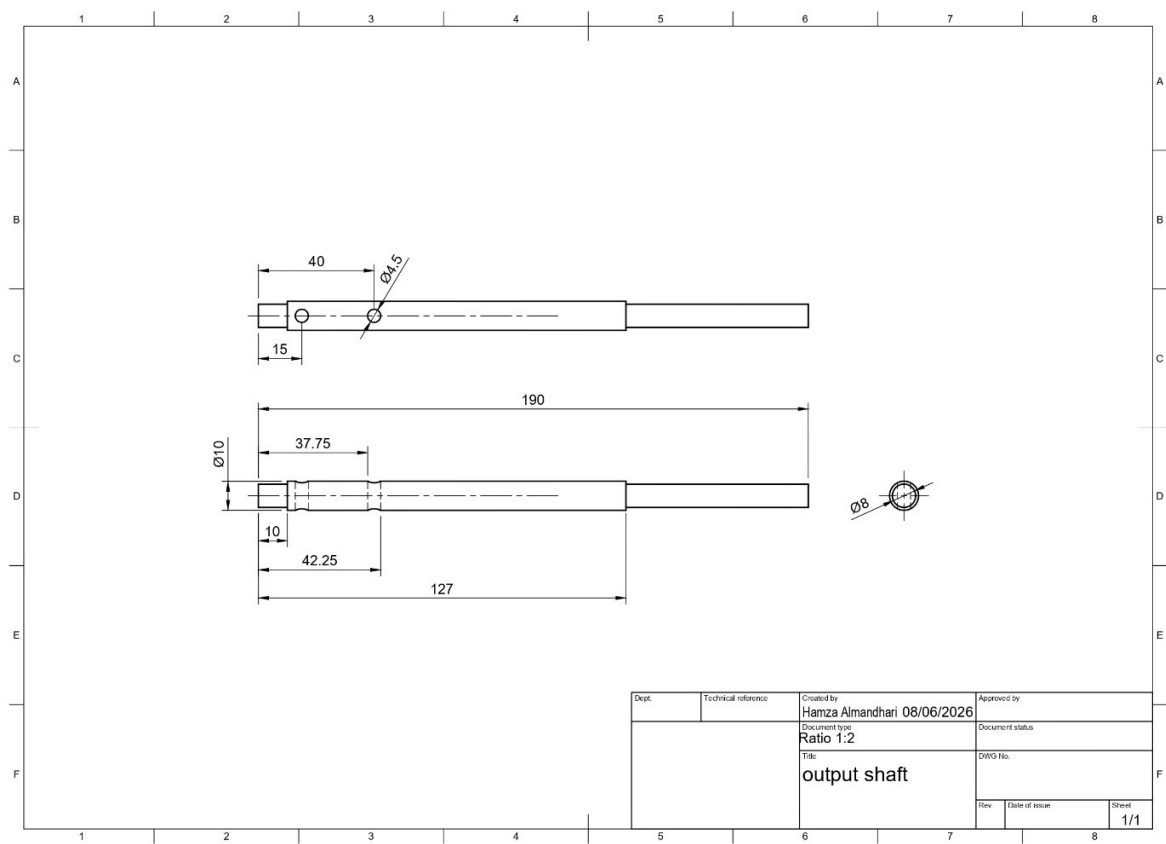
Input Shaft – Aluminum – Lathe Machine



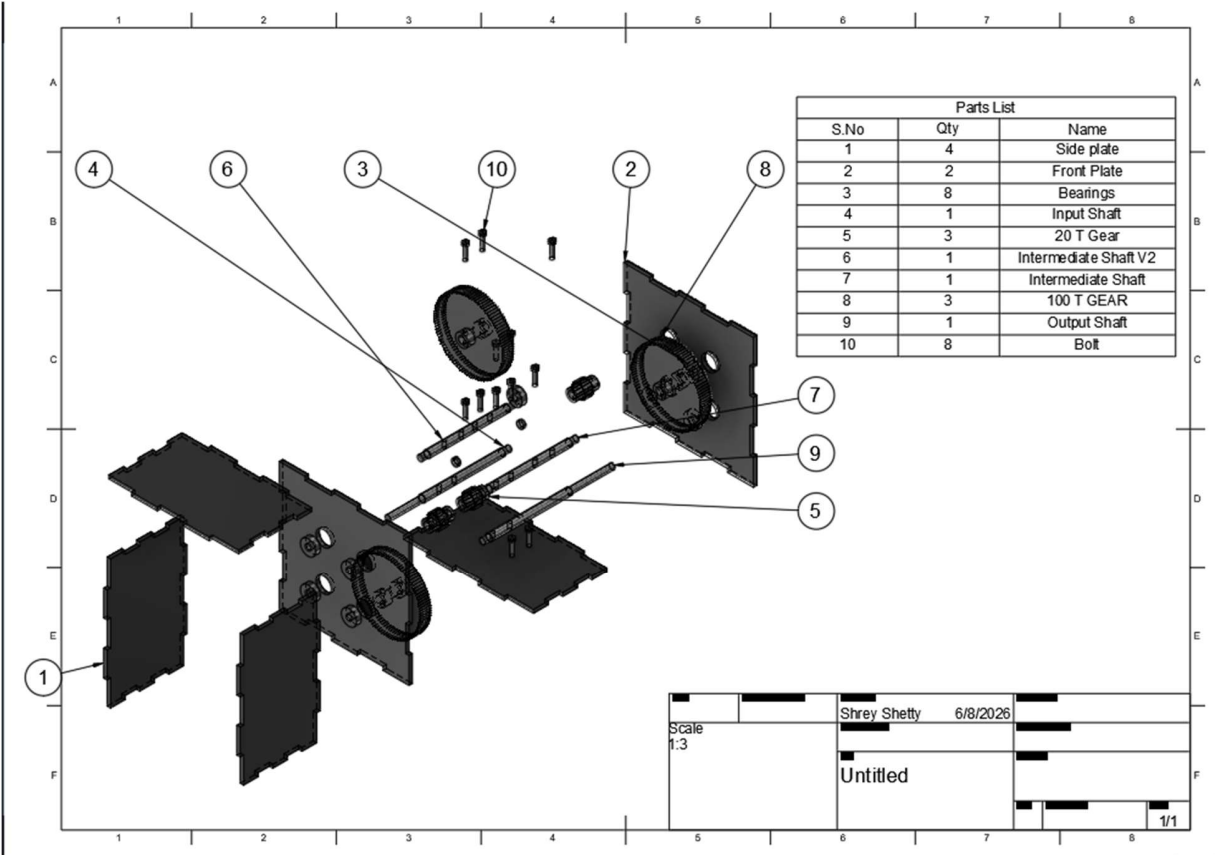
Intermediate Shaft – Aluminum - Lathe Machine



Output Shaft – Aluminum - Lathe Machine



Final Gearbox – Exploded View



BOM (Bill of Materials)

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SUMMARY

This report outlines the analytical and design efforts undertaken for the critical shaft and gearbox assembly as part of the MMMB215 Design and Build project. Stress analysis verified the von Mises stress at the critical section under nominal torque, with yielding checked against DE theory for both Aluminum 6063-O and 3D-printed PLA (100% infill). Fatigue failure was then assessed using the modified Goodman criterion, assuming a machined or cold-drawn surface finish. This yielded a factor of safety that confirms the suitability of the chosen shaft material and geometry. Bearings were selected to meet an operational requirement of a minimum 15,000-hour service life at 98% reliability, with no application factor applied. Every gearbox component was thoroughly detailed in its own engineering drawing, including appropriate tolerance grades, fit notations, and stress concentration relief features. This ensures the drawings are ready for direct workshop use. The complete system assembly drawing, with an exploded view and Bill of Materials, was finished to the same high standard. However, several critical design parameters still need to be finalized before the gearbox can be considered fully specified for manufacturing. The precise lengths of the nut and bolt fasteners must be determined based on the final stacked assembly dimensions to ensure proper thread engagement and clamping force. For the finger joints, exact tolerances need to be set to guarantee correct load transfer and prevent early wear or misalignment during operation. The gearbox housing's dimensional tolerances also require finalization to ensure proper gear mesh, shaft alignment, and bearing seat fits. Lastly, the exact internal orientation of all gearbox components must be verified to confirm clearance, assembly sequence, and the correct torque transmission path. Addressing these outstanding points will be the main focus of the next design phase.